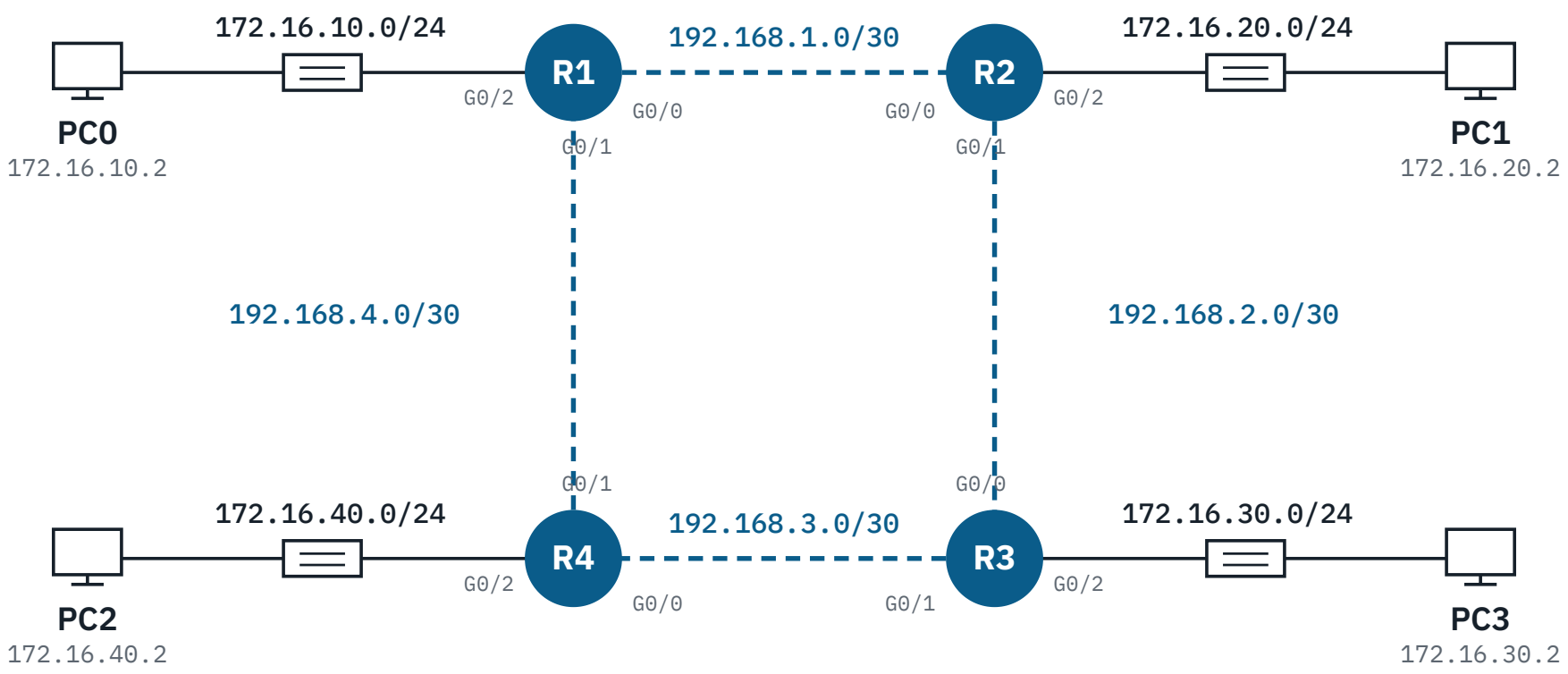
Dynamic routing

RIPv2 on a four router network

Starting from isolated routing tables and ending with full reachability between every LAN. Configured, tested, and reviewed with a security mindset.

ROUTERS	LANS	PROTOCOL	TOOL
4 × Cisco 2911	**4 × /24**	**RIP version 2**	**Packet Tracer**

Four routers in a ring, one LAN each



---- Router to router link, /30 —— LAN, /24, gateway .1 and host .2

ROUTER	INTERFACE	IP ADDRESS	CONNECTS
R1	G0/0	192.168.1.1/30	to R2
R1	G0/1	192.168.4.1/30	to R4
R1	G0/2	172.16.10.1/24	LAN PC0
R2	G0/0	192.168.1.2/30	to R1
R2	G0/1	192.168.2.1/30	to R3
R2	G0/2	172.16.20.1/24	LAN PC1
R3	G0/0	192.168.2.2/30	to R2
R3	G0/1	192.168.3.2/30	to R4
R3	G0/2	172.16.30.1/24	LAN PC3
R4	G0/0	192.168.3.1/30	to R3
R4	G0/1	192.168.4.2/30	to R1
R4	G0/2	172.16.40.1/24	LAN PC2

Each router only knows its own networks

With no routing protocol running, the table on R2 contains only **connected (C)** and **local (L)** routes. R2 has no route to 172.16.10.0/24, so PC1 cannot reach PC0.

R2# show ip route

```
Gateway of last resort is not set

     172.16.0.0/16 is variably subnetted, 2 subnets, 2 masks
C       172.16.20.0/24 is directly connected, GigabitEthernet0/2
L       172.16.20.1/32 is directly connected, GigabitEthernet0/2
     192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.1.0/30 is directly connected, GigabitEthernet0/0
L       192.168.1.2/32 is directly connected, GigabitEthernet0/0
     192.168.2.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.2.0/30 is directly connected, GigabitEthernet0/1
L       192.168.2.1/32 is directly connected, GigabitEthernet0/1
```

PC1> ping 172.16.10.2

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 172.16.10.2

Pinging 172.16.10.2 with 32 bytes of data:

Reply from 172.16.20.1: Destination host unreachable.
Reply from 172.16.20.1: Destination host unreachable.
Reply from 172.16.20.1: Destination host unreachable.
Reply from 172.16.20.1: Destination host unreachable.

Ping statistics for 172.16.10.2:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

> **Result: destination host unreachable, 100% packet loss.**

Same structure on all four routers

Every router advertises its two transit links and its own LAN. Only the **network** statements change from one router to the next.

R1

```
router rip
 version 2
 no auto-summary
 network 192.168.1.0
 network 192.168.4.0
 network 172.16.10.0
 passive-interface g0/2
```

R2

```
router rip
 version 2
 no auto-summary
 network 192.168.1.0
 network 192.168.2.0
 network 172.16.20.0
 passive-interface g0/2
```

R3

```
router rip
 version 2
 no auto-summary
 network 192.168.2.0
 network 192.168.3.0
 network 172.16.30.0
 passive-interface g0/2
```

R4

```
router rip
 version 2
 no auto-summary
 network 192.168.4.0
 network 192.168.3.0
 network 172.16.40.0
 passive-interface g0/2
```

What each command does

`version 2`

Sends classless updates. Subnet masks travel with each route, and updates go to multicast 224.0.0.9 instead of broadcast.

`no auto-summary`

Keeps the real /30 and /24 subnets. Without it, RIP would summarise 172.16.x.x into a single classful /16 and break routing between the LANs.

`network`

Turns RIP on for every interface inside that network and advertises it to the neighbours.

`passive-interface g0/2`

The LAN side still gets advertised, but no RIP updates are sent toward the PCs. Hosts do not need them, and anyone listening on the LAN should not see the routing table.

Routes learned, traffic flows across the network

PC0 now reaches PC1. The traceroute shows the path RIP selected: **R1, then R2, then PC1**. TTL drops from 128 to 126, one per router crossed.

PC0> ping 172.16.20.2 / tracert 172.16.20.2

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 172.16.20.2

Pinging 172.16.20.2 with 32 bytes of data:

Reply from 172.16.20.2: bytes=32 time<1ms TTL=126
Reply from 172.16.20.2: bytes=32 time<1ms TTL=126
Reply from 172.16.20.2: bytes=32 time=8ms TTL=126
Reply from 172.16.20.2: bytes=32 time<1ms TTL=126

Ping statistics for 172.16.20.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 8ms, Average = 2ms

C:\>tracert 172.16.20.2

Tracing route to 172.16.20.2 over a maximum of 30 hops:

  1    0 ms      0 ms      0 ms      172.16.10.1
  2    0 ms      0 ms      1 ms      192.168.1.2
  3    0 ms      0 ms      0 ms      172.16.20.2

Trace complete.
```

> **Result: 4 of 4 replies, 0% loss, trace complete in 3 hops.**

What I take away from this lab

What works

- Routes converge on their own, with no static configuration.
- In a ring, there are two equal cost paths to the opposite LAN, and RIP keeps both.
- Hop count makes path selection easy to predict.

The security gap

- No authentication is configured. Any device speaking RIP on a transit link could inject false routes.
- That allows traffic redirection or black holing.
- Passive interfaces help, but they are not a defence on their own.

Next steps

1. Add MD5 authentication with a key chain.
2. Rebuild the same topology with OSPF and compare.
3. Simulate a rogue router, then detect and block it.

GET THE LAB

The Packet Tracer file (.pkt) is on my GitHub repository. Link in the post description.